

Figure 1: The installation prompt

Now, let's install Tripwire. Being a SourceForge project, you can download it from <http://sourceforge.net/projects/tripwire>. The easiest way to install on the Ubuntu or Backtrack distros is to run `apt-get install tripwire`. Figure 1 shows the first screen of installation, mentioning what Tripwire does, and the security required around the site key.

Running Tripwire

Once the installation is complete, it's important to understand the files created by the tool, and the command set. While the man pages explain everything in detail, the table on the right depicts four commands that every administrator must know, plus three Tripwire user manual pages, which explain each component in detail, with various command-line options.

While configuring Tripwire, it is critical to understand its policies. After the default installation, all administrators would want to tune the tool based on their network's scenario. Each object in the policy file is associated with a property mask, which describes what changes to the file or directory Tripwire should monitor, and which ones can safely be ignored. This fine-tuning helps the systems administrator to closely control how the tool should check the integrity of a file system. Policy configuration is usually a one-time task, and needs to be updated only if the infrastructure or the configuration changes drastically.

A policy sets the rules in the format `object_name -> property_mask`. Here, the object name is a fully qualified path to a file or directory; the `property_mask` specifies what property to examine or skip. For instance, if you want the tool to scan the entire `/var/log` folder and files, except the `mail.log` file, the policy would be set as follows:

```
/var/log -> S(ReadOnly);
! /var/log/mail.log;
```

Please refer to the man pages of `twpolicy` to understand the settings. Figure 2 shows a screen where Tripwire is run in interactive mode, which also shows the locations of the policy and configuration files, which can be tweaked as per your need.

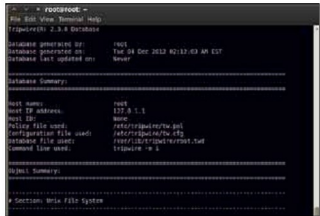


Figure 2: Tripwire in interactive mode

Tripwire Command	Usage
twadmin	Creates and maintains configuration and key files.
twprint	Prints Tripwire database and report files in plain-text.
tripwire	Creates and maintains baseline database. Checks file system's integrity against that database.
siggen	Utility that displays hash values for files. Required for scriptomatic checks.
man twfiles	Explains various files of Tripwire and default locations.
man twconfig	Explains configuration files that enhance usage.
man twpolicy	Explains policies that Tripwire should follow while scanning.

As we saw, Tripwire is a complete command-line-operated host-based intrusion prevention tool. Administrators can run it using a `cron` job, and `grep` and parse its output to generate alerts. In a mid-sized IT infrastructure, by using proper user permissions, Tripwire can scan all servers and save logs on a centralised database for future attack analysis, or for compliance reasons. Large organisations can make this tool part of their code-release management cycle. This can help avoid situations where code inadvertently overwrites a system or critical application file, thus improving post-installation stability. The tool can be used to ensure integrity of plain-text databases too, which are usually found on older Linux systems. 

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